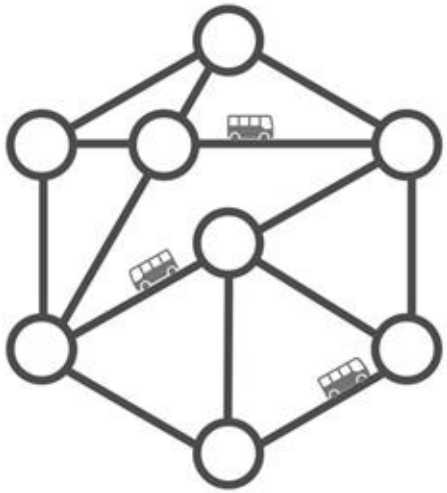
Journey Planner Project

Oscar Goudet
Sayantan Biswas
Mathis Magnin
Sajal Chaurasia
Fletcher Collis

Presentation plan

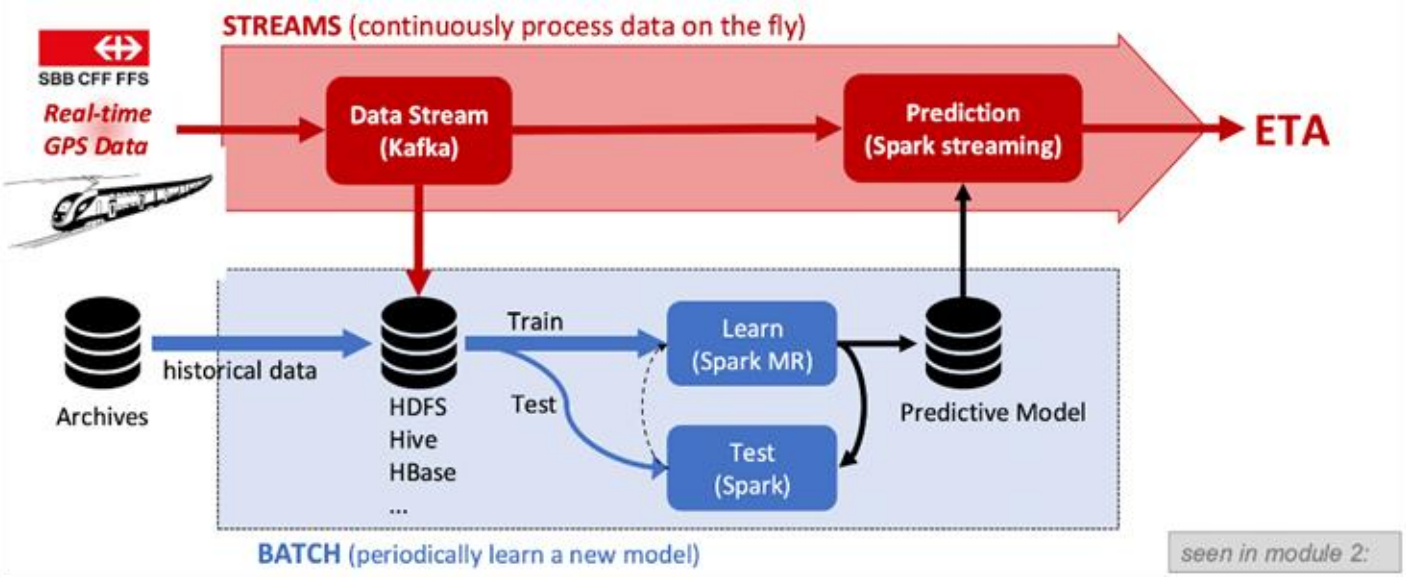
1. Introduction
2. Data Structure
3. Routing algorithm
4. Modeling of the delays
5. Validation method
6. Visualization and user interface



Introduction

Introduction

Goal : Trying to be as realistic as possible in terms of data availability



→ using data after the query time we consider as data leakage

Respecting query time data boundaries

This project hypothesis

We implemented two different data pipelines

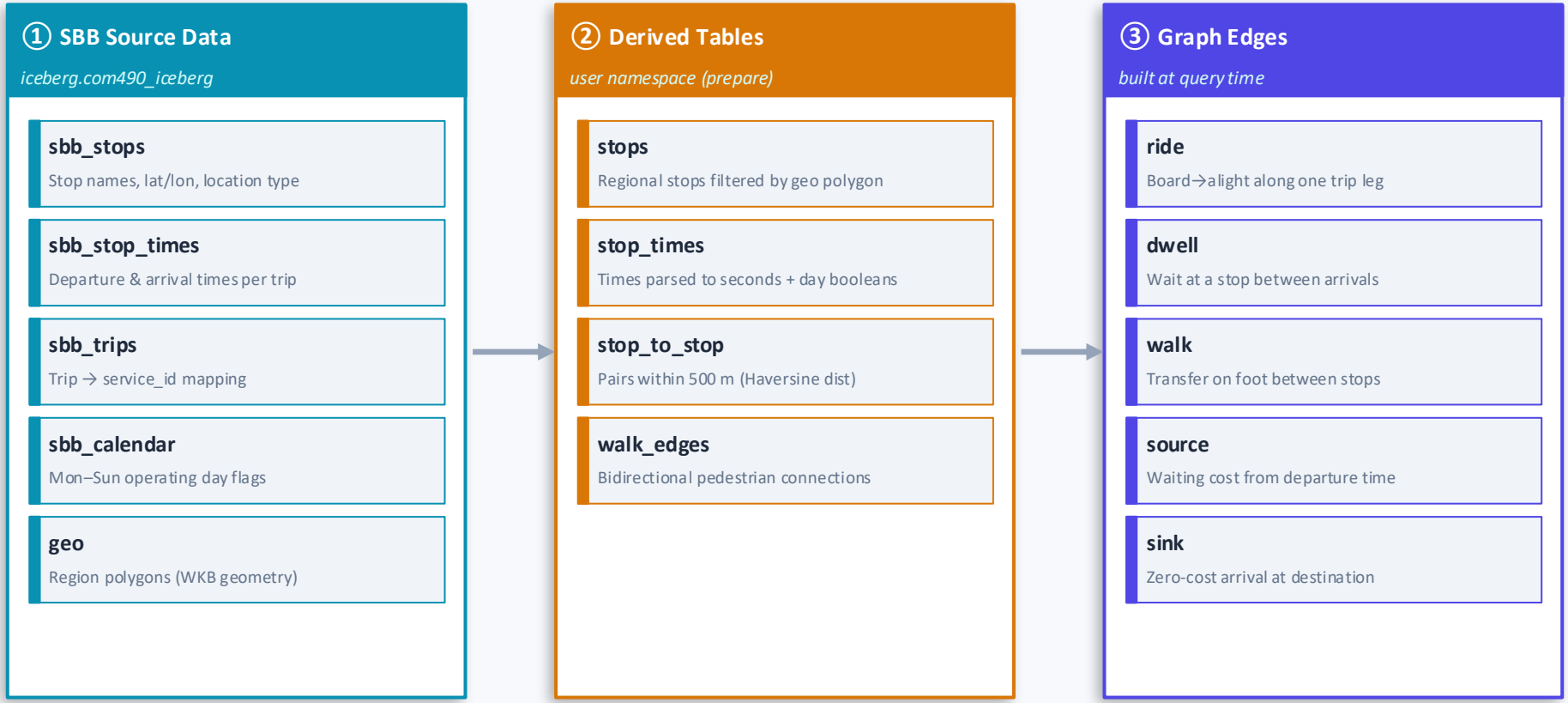
- historical data we consider all data up until last month of data
- current data we consider all data up to the query time.



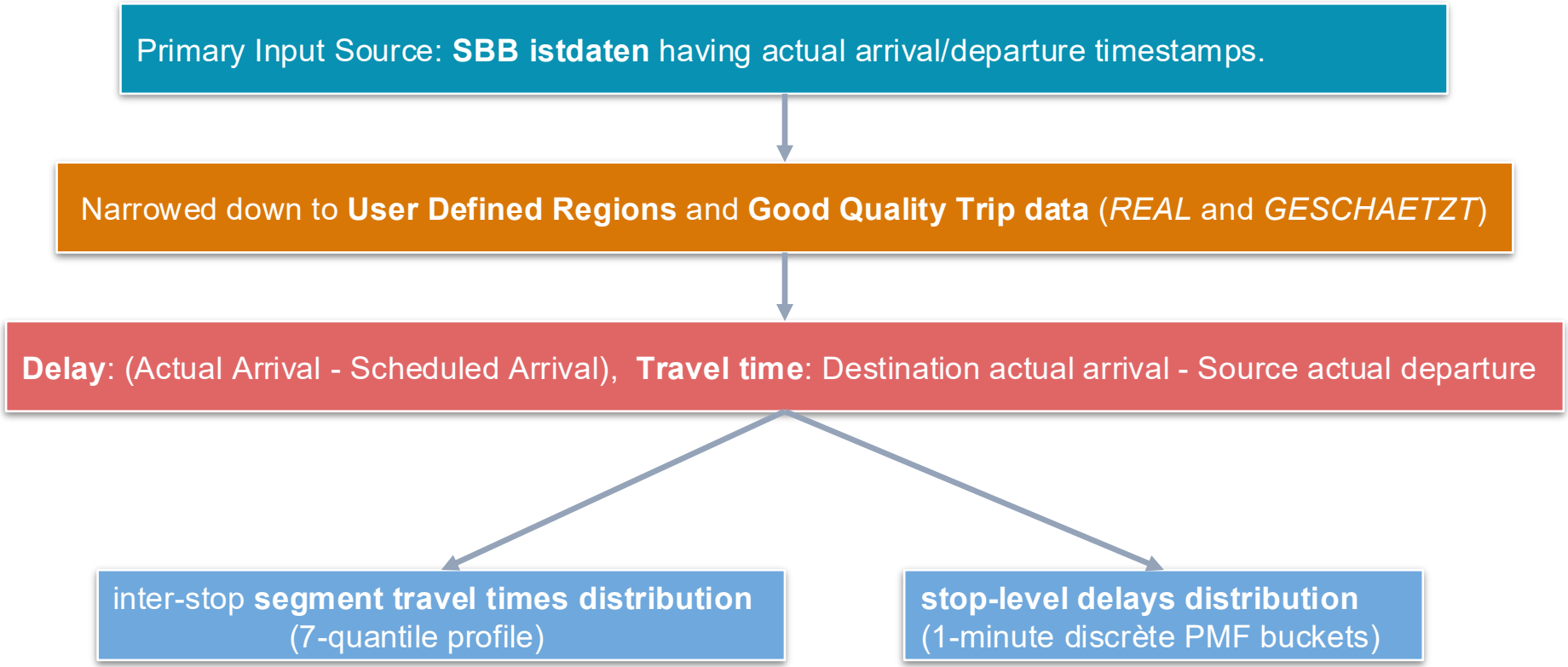
Data Structure

Routing Data Pipeline

How graph edges are built from raw SBB GTFS data



Delay and Travel Times Modelling Data Pipeline



Routing Algorithm

Chosen Routing Algorithm

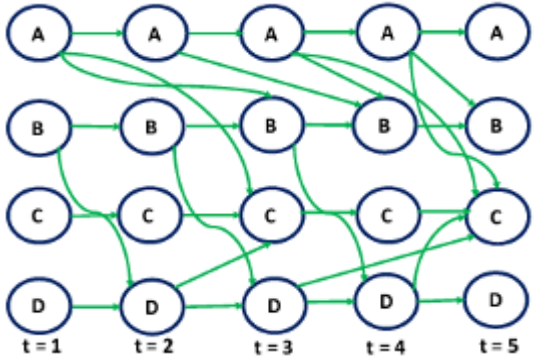
Time Expanded Graph

- Use the timetable to time-expand the graph where each trip between two stops becomes an edge with its own dep. and arr. time

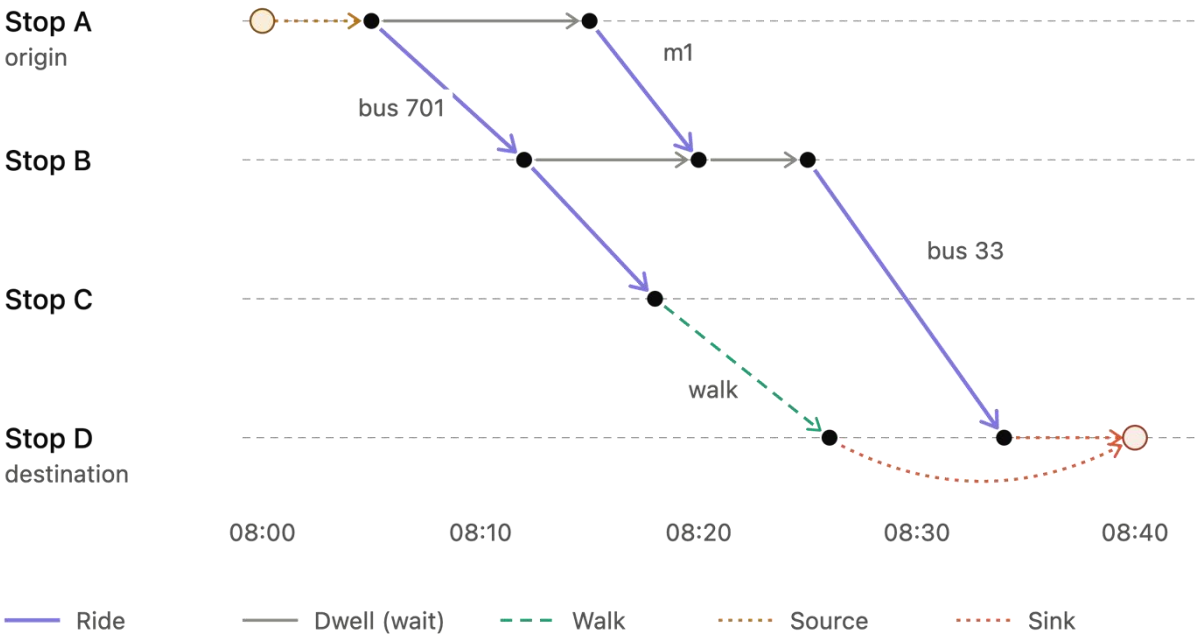
Backwards Dijkstras Algorithm

- Adaptation of classical algorithm to anchor at the arrival and search backwards in time to find the latest possible departure
- Modified to give back the k-best paths (keeps running instead of stopping at best)





Example of a time-expanded graph



Chosen Routing Algorithm

Transfer Buffer

- Variable buffer added at transfers as to give more conservative solutions

Backwards Sweep

- Variable arrival time at destination as to give different alternative routes





→ **Output = collection of paths from origin to destination that respect the arrival deadline**

Delay Modeling

Modelling delays — empirical PMFs, conditioned by hour

Stop delay PMF

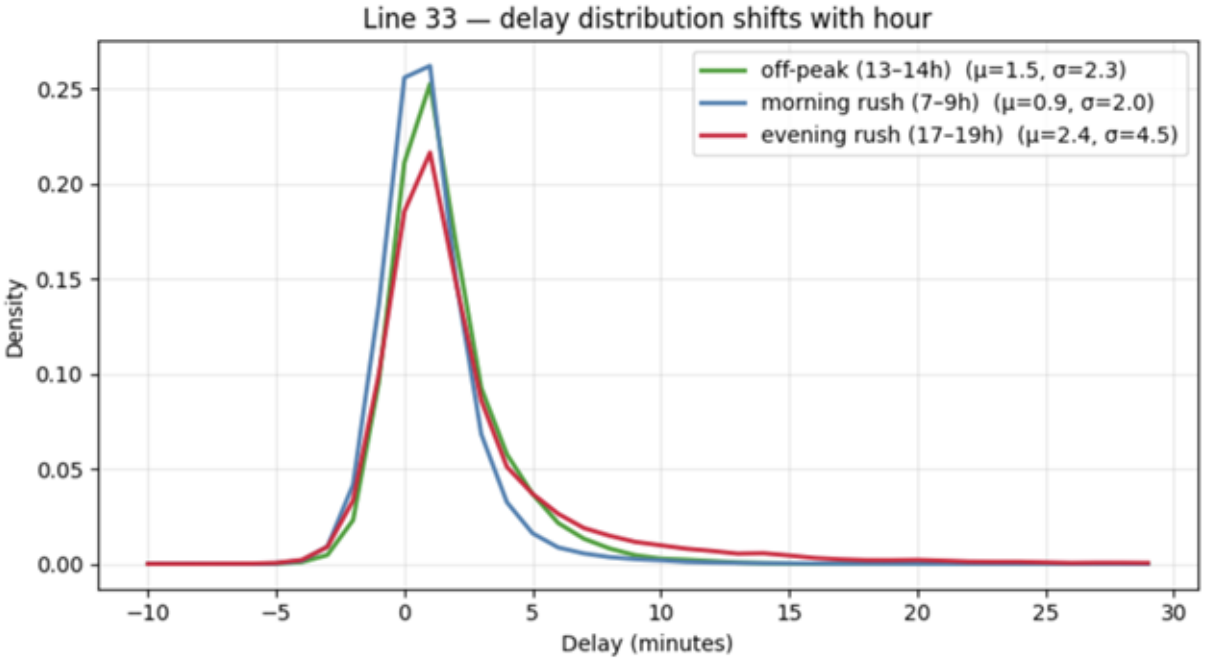
-**no live observation** of the bus yet.

-Sample a historical delay at the scheduled time.

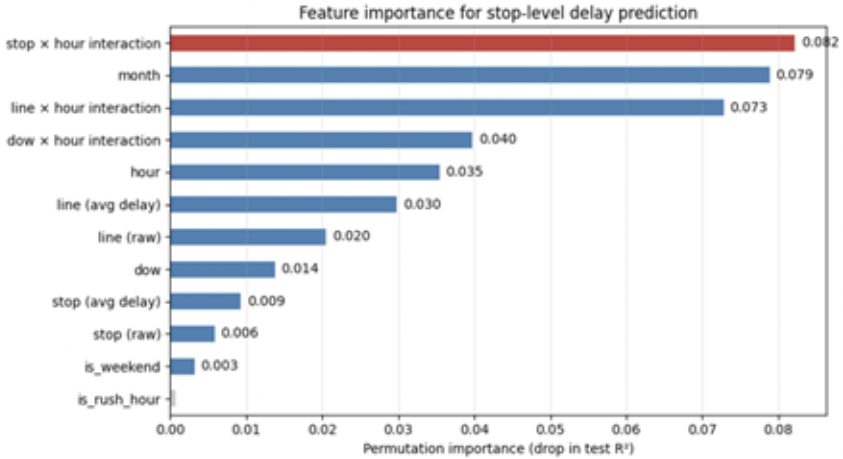
Segment travel time

-bus is **already running** and we see its position.

-Forecast its travel time from current position to the next stop.



Validated features · fallback · holdout



Final conditioning

`[dow] [line] [stop] [hour]`

Top features are hour-interactions
→ conditioning anchors on hour

month strong but dropped → *keep cell sample size > 50*

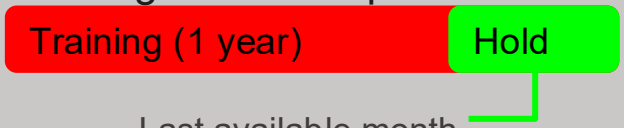
Three-tier fallback (min 50 obs)

`[dow] [line] [stop] [hour]`

`[line] [stop] [hour]`

`[stop] [hour]`

Training / holdout split



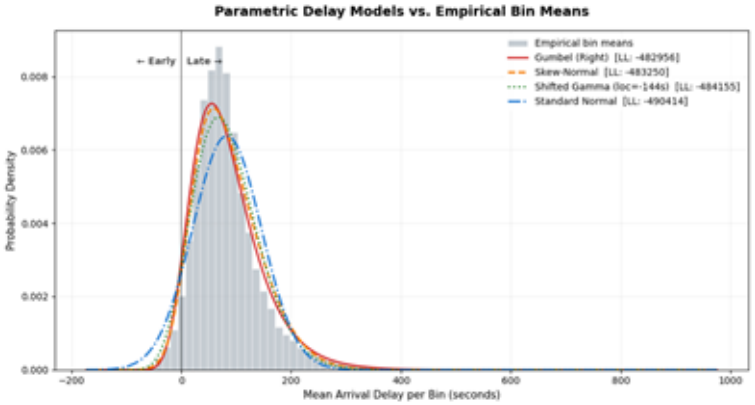
Model never sees the day it's tested on

Validation method

Parametric Model & Validation

We Test parametric models as it provides faster inference and better generalization in production. Among all skewed distributions tested, Gumbel best captures our empirical data.

Model	MAE (sec)
Stop-level Discrete Median	98.04
Gumbel MAE	97.49
Global Gumbel	100.19



Threshold	Discrete MAE	Gumbel MAE	Gumbel Reg
-120	0.02577	0.03161	0.02600
0	0.14323	0.08210	0.11083
60	0.12928	0.08020	0.13192
180	0.05560	0.06638	0.08619
300	0.03264	0.04243	0.04811
600	0.01336	0.01441	0.01649

User Interface

Visualization

